UNDER A FLAT LOCAL MAP INDUCING AN ISOMORPHISM ON RESIDUE FIELDS THE INJECTIVE HULLS AGREE

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Allowed standard prerequisites (editorial).

- The statement that an injective module over the source stays injective over the target under a flat map. Invoked inside the counted proof (source lines 1072-1074). Makes the hull of the target injective over the source.
- The criterion for an essential extension of a module to be its injective hull through the quotients by the powers of the maximal ideal. Invoked at the end of the counted proof (source lines 1075-1079). Upgrades the comparison map to an isomorphism.

Source reference. Original lemma lemma-injective-flat: sources/stacks-dualizing/pinned-original-source.tex, source lines 410–410. Navigation pointer only: the counted text refers to the Stacks item lemma-injective-flat, which is an existing labeled result of this same chapter and is not part of this card. The pointer carries the target's own label as its number and its definition line as its span; its content is not reproduced here.

Source reference. Original lemma lemma-union-artinian: sources/stacks-dualizing/pinned-original-source.tex, source lines 1042–1042. Navigation pointer only: the counted text refers to the Stacks item lemma-union-artinian, which is an existing labeled result of this same chapter and is not part of this card. The pointer carries the target's own label as its number and its definition line as its span; its content is not reproduced here.

Lemma 0.1. *Let $R \to S$ be a flat local homomorphism of local Noetherian rings such that $R/\mathfrak{m}_R \cong S/\mathfrak{m}_R S$. Then the injective hull of the residue field of R is the injective hull of the residue field of S.*

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Proof. Note that $\mathfrak{m}_R S = \mathfrak{m}_S$ as the quotient by the former is a field. Set $\kappa = R/\mathfrak{m}_R = S/\mathfrak{m}_S$. Let E_R be the injective hull of κ over R. Let E_S be the injective hull of κ over S. Observe that E_S is an injective R-module by Lemma lemma-injective-flat. Choose an extension $E_R \to E_S$ of the identification of residue fields. This map is an isomorphism by Lemma lemma-union-artinian because $R \to S$ induces an isomorphism $R/\mathfrak{m}_R^n \to S/\mathfrak{m}_S^n$ for all n. $\square$